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CPSC-482-02

## **1. Project Overview and Goals**

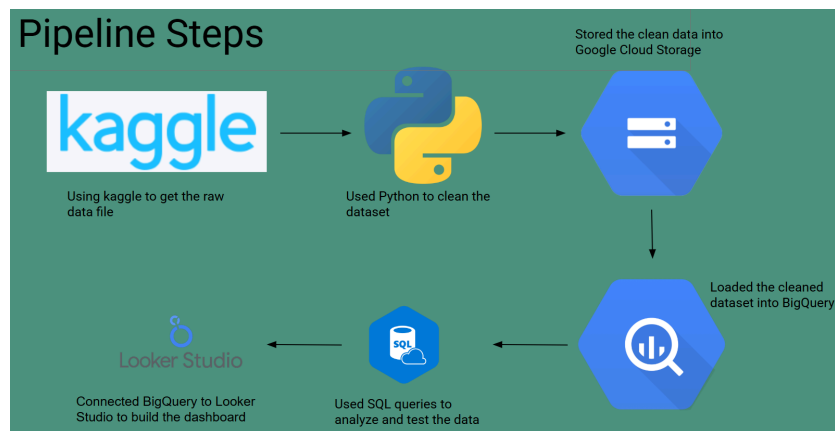
Since I personally very like to play video games and use Steam often, so I found this dataset and made this project. The goal of my project is to build a video game analytics pipeline using Google Cloud and Big Query. The project uses a Steam game analytics dataset from Kaggle and using python to let raw CSV file become cleaned data, also tested by SQL query, and build a dashboard using Looker Studio. The main purpose of this project is to help people that do not play video games have a better understanding of video game market trends. I focus on questions such as what genres are most popular, how game prices are distributed, how many games are free or paid, and to see which game is the most popular right now, using concurrent.

## **2. Dataset Description**

The dataset used in this project is a Steam game analytics CSV file from Kaggle. It is a structured dataset with about 220,000 records and 28 columns. The dataset includes many useful attributes, but not every one is useful, so the attributes that I use is:

- game name
- primary genre
- price
- maybe free-to-play status / paid status
- total game count
- concurrent users
- publisher
- reviews
- release date

### 3. Pipeline Workflow and Services Used



First, I downloaded the raw Steam game analytics CSV file from Kaggle. After I got the data file I used Python to clean the dataset. The cleaning took most of the time for this project. After the cleaning, I uploaded the cleaned dataset into Google Cloud Storage. I created a new Bucket for my video game clean data file. Next, I loaded the cleaned data file from Google Cloud Storage into BigQuery. After the file is in BigQuery then I start writing SQL queries in BigQuery to test the data to see if there were more data that need to be clean or not and also generate useful summaries. Finally, I connected BigQuery to Looker Studio and created a dashboard to visualize the results.

Buckets [+ Create](#) [Refresh](#)

Filter Filter buckets

| <input type="checkbox"/> | Name <a href="#">↑</a>                   | Created                 | Location type | Location | Default storage class <a href="#">?</a> | Last modified           | Public access <a href="#">?</a> |
|--------------------------|------------------------------------------|-------------------------|---------------|----------|-----------------------------------------|-------------------------|---------------------------------|
| <input type="checkbox"/> | <a href="#">my-game-analytics-bucket</a> | May 5, 2026, 4:05:13 PM | Multi-region  | us       | Standard                                | May 5, 2026, 4:05:13 PM | Not public                      |

final-project-495422 / Datasets / game\_analytics / Tables / cleaned\_games

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Filter Enter property name or value

| <input type="checkbox"/> | Field name       | Type    | Mode     | Description |
|--------------------------|------------------|---------|----------|-------------|
| <input type="checkbox"/> | appid            | INTEGER | NULLABLE | -           |
| <input type="checkbox"/> | name             | STRING  | NULLABLE | -           |
| <input type="checkbox"/> | genre_primary    | STRING  | NULLABLE | -           |
| <input type="checkbox"/> | developer        | STRING  | NULLABLE | -           |
| <input type="checkbox"/> | publisher        | STRING  | NULLABLE | -           |
| <input type="checkbox"/> | publisher_tier   | STRING  | NULLABLE | -           |
| <input type="checkbox"/> | price_initial    | FLOAT   | NULLABLE | -           |
| <input type="checkbox"/> | price_final      | FLOAT   | NULLABLE | -           |
| <input type="checkbox"/> | discount_percent | FLOAT   | NULLABLE | -           |
| <input type="checkbox"/> | owners_min       | INTEGER | NULLABLE | -           |
| <input type="checkbox"/> | owners_max       | INTEGER | NULLABLE | -           |
| <input type="checkbox"/> | owners_midpoint  | INTEGER | NULLABLE | -           |
| <input type="checkbox"/> | positive_reviews | INTEGER | NULLABLE | -           |
| <input type="checkbox"/> | negative_reviews | INTEGER | NULLABLE | -           |

## 4. Data Processing Steps

I used Python and pandas locally to clean the raw `game_analytics.csv` file. First, I removed rows that were missing important fields, such as `appid` and `name`, because each game needs an ID and a name. Then, I take care of the missing values: Text columns, such as `genre`, `developer`, `publisher`, and `tags`, will have the word "Unknown". Numeric columns, such as `price`, `reviews`, `owners`, and `concurrent users`, were filled with 0. Boolean columns, such as `is_free`, `is_early_access`, and `steam_deck`, were filled with False. After that, I changed the data types of different columns: IDs and count columns were changed to integers, price columns were changed to floats, and `release_date` was changed into a date format. I also created new columns, such as `release_year`, `total_reviews`, `review percentages`, `game_type`, `has_discount`, and `has_achievements`. Finally, I removed duplicate games by cleaning the game names and keeping the record with the highest reviews, peak concurrent users, and estimated owners. I also removed long PC requirement columns that were not needed for my dashboard. The final cleaned file was saved as `cleaned_game_analytics.csv`.

```
1 import pandas as pd
2
3 df = pd.read_csv("game_analytics.csv")
4
5 print("Original shape:", df.shape)
6
7 df = df.dropna(subset=["appid", "name"])
8
9 text_columns = [
10     "genre_primary",
11     "developer",
12     "publisher",
13     "publisher_tier",
14     "tags",
15     "pc_req_min",
16     "pc_req_rec"
17 ]
18
19 for col in text_columns:
20     df[col] = df[col].fillna("Unknown")
21
22 numeric_columns_zero = [
23     "price_initial",
24     "price_final",
25     "discount_percent",
26     "owners_min",
27     "owners_max",
28     "owners_midpoint",
29     "positive_reviews",
30     "negative_reviews",
31     "ccu",
32     "peak_ccu",
33     "required_age",
34     "languages_count",
35     "achievement_count",
36     "dlc_count"
37 ]
38
39 for col in numeric_columns_zero:
40     df[col] = df[col].fillna(0)
```

```

41
42 ✓ boolean_columns = [
43     "is_early_access",
44     "is_free",
45     "steam_deck"
46 ]
47
48 ✓ for col in boolean_columns:
49     df[col] = df[col].fillna(False)
50
51     df["controller_support"] = df["controller_support"].fillna("Unknown")
52
53     df["appid"] = df["appid"].astype(int)
54
55 ✓ int_columns = [
56     "owners_min",
57     "owners_max",
58     "owners_midpoint",
59     "positive_reviews",
60     "negative_reviews",
61     "ccu",
62     "peak_ccu",
63     "required_age",
64     "languages_count",
65     "achievement_count",
66     "dlc_count"
67 ]
68
69 ✓ for col in int_columns:
70     df[col] = df[col].astype(int)
71
72 ✓ float_columns = [
73     "price_initial",
74     "price_final",
75     "discount_percent"
76 ]

```

```

77
78 ✓ for col in float_columns:
79     df[col] = df[col].astype(float)
80
81     df["release_date"] = pd.to_datetime(df["release_date"], errors="coerce")
82
83     df["release_year"] = df["release_date"].dt.year
84
85     df["release_date"] = df["release_date"].dt.strftime("%Y-%m-%d")
86
87     df["total_reviews"] = df["positive_reviews"] + df["negative_reviews"]
88
89 ✓ df["positive_review_percent"] = df.apply(
90     lambda row: round((row["positive_reviews"] / row["total_reviews"]) * 100, 2)
91     if row["total_reviews"] > 0 else 0,
92     axis=1
93 )
94
95 ✓ df["negative_review_percent"] = df.apply(
96     lambda row: round((row["negative_reviews"] / row["total_reviews"]) * 100, 2)
97     if row["total_reviews"] > 0 else 0,
98     axis=1
99 )
100

```

```

100
101 df["game_type"] = df["is_free"].apply(
102     lambda x: "Free-to-play" if x == True else "Paid"
103 )
104
105 df["has_discount"] = df["discount_percent"].apply(
106     lambda x: True if x > 0 else False
107 )
108
109 df["has_achievements"] = df["achievement_count"].apply(
110     lambda x: True if x > 0 else False
111 )
112
113 before_dedup = df.shape[0]
114
115 df["name_clean"] = df["name"].astype(str).str.strip().str.lower()
116
117 df = df.sort_values(
118     by=["total_reviews", "peak_ccu", "owners_midpoint"],
119     ascending=[False, False, False]
120 )
121
122 df = df.drop_duplicates(subset=["name_clean"], keep="first")
123
124 df = df.drop(columns=["name_clean"])
125
126 after_dedup = df.shape[0]
127
128 print("Rows before duplicate removal:", before_dedup)
129 print("Rows after duplicate removal:", after_dedup)
130 print("Duplicate rows removed:", before_dedup - after_dedup)
131
132 df = df.drop(columns=["pc_req_min", "pc_req_rec"], errors="ignore")
133
134 df.to_csv("cleaned_game_analytics.csv", index=False, encoding="utf-8")
135
136 print("Cleaned shape:", df.shape)
137 print("Cleaned file saved as cleaned_game_analytics.csv")
138 print(df.head())

```

## 5. Results Obtained

After the data was loaded into BigQuery, I used SQL queries to generate useful summaries from the dataset. The main results included:

- number of games by genre
- price distribution
- total number of games
- Number of games by publisher
- Free-to-play vs paid games
- Top games by peak concurrent users
- Average peak CCU by genre
- Review pattern by genre

The result of these queries show that there are 210,077 games in total. And the genre result showed that Action has the largest genre with 83,462 games, followed by Casual and Adventure. This means that Action is the most popular game genre right now. The publisher result showed

the top publishers after filtering out unknown publishers and games with very few reviews. I also compared free-to-play and paid games. I found out that most of the games are paid, which means that you have to pay some money to play video right now. For price distribution, most games were either free or in the \$0.01–\$5 range. Which means that most of the game after a few years they will become chipper. I also checked peak concurrent users and review patterns by genre. The top game by peak users is Counter-Strike: Global Offensive, which means that this game is the most popular game for the Steam video game market.

| Row | total_games |
|-----|-------------|
| 1   | 210077      |

| Row | genre_primary         | game_count |
|-----|-----------------------|------------|
| 1   | Action                | 83462      |
| 2   | Casual                | 38521      |
| 3   | Adventure             | 38180      |
| 4   | Indie                 | 21895      |
| 5   | Simulation            | 8640       |
| 6   | RPG                   | 6085       |
| 7   | Strategy              | 3251       |
| 8   | Racing                | 1725       |
| 9   | Free To Play          | 1504       |
| 10  | Unknown               | 1394       |
| 11  | Design & Illustration | 1364       |
| 12  | Animation & Modeling  | 907        |

| Row | publisher                  | game_count |
|-----|----------------------------|------------|
| 1   | Paradox Interactive        | 184        |
| 2   | Kagura Games               | 171        |
| 3   | Square Enix                | 170        |
| 4   | Ubisoft                    | 162        |
| 5   | THQ Nordic                 | 146        |
| 6   | Devolver Digital           | 135        |
| 7   | Sekai Project              | 126        |
| 8   | Dovetail Games - Trains    | 124        |
| 9   | SEGA                       | 124        |
| 10  | Fulqrum Publishing         | 120        |
| 11  | Conglomerate 5             | 119        |
| 12  | BANDAI NAMCO Entertainment | 107        |
| 13  | Strategy First             | 106        |
| 14  | Electronic Arts            | 104        |
| 15  | Nacon                      | 100        |
| 16  | KOEI TECMO GAMES CO., LTD. | 100        |
| 17  | PLAYISM                    | 97         |
| 18  | SA Industry                | 94         |
| 19  | My Way Games               | 94         |
| 20  | Microids                   | 93         |

| Row | game_type ▾  | game_count ▾ | total_positive_rev... | total_negative_re... | total_reviews ▾ | weighted_positiv... |
|-----|--------------|--------------|-----------------------|----------------------|-----------------|---------------------|
| 1   | Free-to-play | 4750         | 26216974              | 6238399              | 32455373        | 80.78               |
| 2   | Paid         | 34578        | 95935109              | 13291026             | 109226135       | 87.83               |

| Row | price_range ▾  | game_count ▾ |
|-----|----------------|--------------|
| 1   | \$0.01 - \$5   | 79245        |
| 2   | Free           | 63744        |
| 3   | \$5.01 - \$10  | 33887        |
| 4   | \$10.01 - \$20 | 23638        |
| 5   | \$20.01 - \$40 | 7422         |
| 6   | Over \$40      | 2141         |

| Row | name ▾                           | genre_primary ▾ | publisher ▾                      | peak_ccu ▾ |
|-----|----------------------------------|-----------------|----------------------------------|------------|
| 1   | Counter-Strike: Global Offensive | Action          | Valve                            | 1013936    |
| 2   | Dota 2                           | Action          | Valve                            | 623941     |
| 3   | PUBG: BATTLEGROUNDS              | Action          | KRAFTON, Inc.                    | 314682     |
| 4   | Bongo Cat                        | Casual          | Irox Games                       | 164168     |
| 5   | ELDEN RING NIGHTREIGN            | Action          | FromSoftware, Inc., Bandai Na... | 163599     |
| 6   | Rust                             | Action          | Facepunch Studios                | 143870     |
| 7   | Apex Legends                     | Action          | Electronic Arts                  | 124262     |
| 8   | Marvel Rivals                    | Action          | NetEase Games                    | 106611     |
| 9   | Wallpaper Engine                 | Casual          | Wallpaper Engine Team            | 91184      |
| 10  | DELTARUNE                        | Indie           | tobyfox                          | 83936      |
| 11  | Call of Duty: Modern Warfare II  | Action          | Activision                       | 67419      |
| 12  | NARAKA: BLADEPOINT               | Action          | NetEase Games Global             | 66954      |
| 13  | Baldur's Gate 3                  | Adventure       | Larian Studios                   | 54771      |
| 14  | HELLDIVERS 2                     | Action          | PlayStation Publishing LLC       | 53399      |
| 15  | Tom Clancy's Rainbow Six Siege   | Action          | Ubisoft                          | 51847      |
| 16  | Stardew Valley                   | Indie           | ConcernedApe                     | 50662      |
| 17  | War Thunder                      | Action          | Gaijin Network Ltd               | 50478      |
| 18  | Dead by Daylight                 | Action          | Behaviour Interactive Inc.       | 44886      |
| 19  | Team Fortress 2                  | Action          | Valve                            | 43819      |
| 20  | EA SPORTS FC 25                  | Simulation      | Electronic Arts                  | 42032      |

| Row | genre_primary ▾       | game_count ▾ | avg_peak_ccu ▾ |
|-----|-----------------------|--------------|----------------|
| 1   | Software Training     | 3            | 1127.67        |
| 2   | Massively Multiplayer | 56           | 927.04         |
| 3   | Action                | 6434         | 710.72         |
| 4   | Audio Production      | 30           | 543.63         |
| 5   | Simulation            | 638          | 395.0          |
| 6   | Game Development      | 3            | 357.67         |
| 7   | RPG                   | 561          | 339.28         |
| 8   | Animation & Modeling  | 89           | 305.62         |
| 9   | Strategy              | 452          | 289.55         |
| 10  | Racing                | 152          | 252.34         |
| 11  | Utilities             | 86           | 233.58         |
| 12  | Early Access          | 5            | 221.8          |
| 13  | Indie                 | 1743         | 203.52         |
| 14  | Casual                | 2338         | 126.87         |
| 15  | Adventure             | 3629         | 92.18          |
| 16  | Unknown               | 78           | 83.08          |
| 17  | Free To Play          | 126          | 63.97          |
| 18  | Sports                | 41           | 47.93          |
| 19  | Design & Illustration | 31           | 25.55          |
| 20  | Nudity                | 6            | 24.83          |

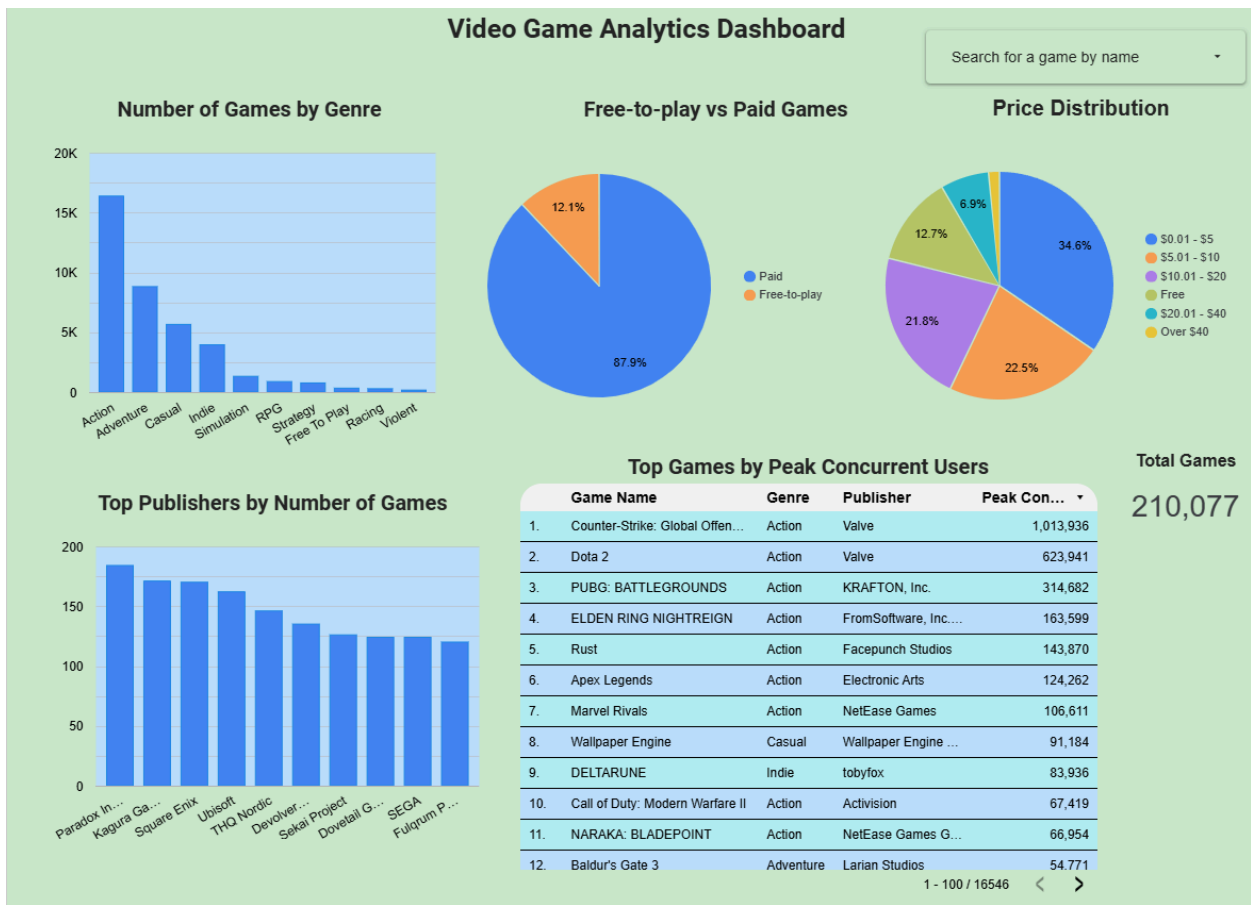
Query results

| Job information |                       | Results      | Visualization         | JSON                 | Execution details | Execution graph     |  |
|-----------------|-----------------------|--------------|-----------------------|----------------------|-------------------|---------------------|--|
| Row             | genre_primary ▾       | game_count ▾ | total_positive_rev... | total_negative_re... | total_reviews ▾   | weighted_positiv... |  |
| 1               | Movie                 | 4            | 10399                 | 120                  | 10519             | 98.86               |  |
| 2               | Documentary           | 1            | 424                   | 19                   | 443               | 95.71               |  |
| 3               | Episodic              | 2            | 480                   | 35                   | 515               | 93.2                |  |
| 4               | Audio Production      | 40           | 97633                 | 8212                 | 105845            | 92.24               |  |
| 5               | Animation & Modeling  | 142          | 166692                | 14505                | 181197            | 91.99               |  |
| 6               | Indie                 | 3949         | 7583182               | 707763               | 8290945           | 91.46               |  |
| 7               | Casual                | 5653         | 7290792               | 712800               | 8003592           | 91.09               |  |
| 8               | Early Access          | 12           | 12593                 | 1258                 | 13851             | 90.92               |  |
| 9               | Adventure             | 8808         | 12835447              | 1617601              | 14453048          | 88.81               |  |
| 10              | Racing                | 277          | 875422                | 111990               | 987412            | 88.66               |  |
| 11              | Strategy              | 760          | 2767703               | 383206               | 3150909           | 87.84               |  |
| 12              | Unknown               | 150          | 373606                | 55140                | 428746            | 87.14               |  |
| 13              | RPG                   | 874          | 4445550               | 713415               | 5158965           | 86.17               |  |
| 14              | Game Development      | 3            | 7131                  | 1192                 | 8323              | 85.68               |  |
| 15              | Action                | 16374        | 81743663              | 14313660             | 96057323          | 85.1                |  |
| 16              | Utilities             | 127          | 92873                 | 17327                | 110200            | 84.28               |  |
| 17              | Free To Play          | 326          | 219984                | 45125                | 265109            | 82.98               |  |
| 18              | Gore                  | 18           | 7104                  | 1530                 | 8634              | 82.28               |  |
| 19              | Massively Multiplayer | 74           | 474406                | 105225               | 579631            | 81.85               |  |
| 20              | Simulation            | 1315         | 3024860               | 681799               | 3706659           | 81.61               |  |



6. Visualizations / Dashboard

I built the final dashboard in Looker Studio using the BigQuery table. As you can see the photo below, the dashboard has a couple charts and one big table. This first chart shows the number of games by genre, which helps identify the most popular game types. It also includes a free-to-play vs paid games pie chart, showing that paid games make up most of the dataset. The one on its right shows a price distribution chart, it shows how games are grouped by price range. This helps show that many games are free or low-priced. Another chart shows the top publishers by number of games, which can let people know which game company publishes most of the game. The dashboard also includes a table for top games by peak concurrent users. This table shows the most popular games right now by showing the concurrent user. You can tell that Counter-Strike: Global Offensive has over a million people playing it now.



## **7. Challenges**

One challenge that I have was cleaning the raw dataset. The dataset had missing values, messy text, and some columns that needed better formatting. This made the cleaning step important before loading the data into BigQuery. Because BigQuery does not allow these messy text loads into the server. I resolved this by using Python to clean missing values, remove messy texts and also delete some columns that I don't need. Another challenge was choosing which results to show in the final dashboard. The dataset had many columns, so there were many possible analysis directions. I solved this by focusing on simple and useful market trends, such as genre count, price distribution, and total game count. So the user can know which kind of the game is most popular right now just by looking at my dashboard.

## **8. Lessons Learned**

I learned how to build a full data analytics pipeline using cloud services. I also learned that data cleaning is one of the most important parts of a data project because raw data often has missing values, messy text, and formatting problems. Before we use this data, we have to make sure all these problems are fixed, otherwise this problem will cause some error while I am working on the project. And I also noticed that Google Cloud Storage is useful for storing the dataset, you can just simply upload the file that you want to work on and use BigQuery for querying and analyzing the data. The last thing that I learned is that the dashboards should be simple and easy to understand. For my dashboard, it's design is good, but sometimes it's not very easy for users to understand it. So I should focus on the most useful insights so users can understand the data quickly.

## **9. Next Steps**

If I had more time, I will look for more game datasets, such as mobile games or games that are not available on Steam. This may help the project show a bigger picture of the video game market instead of only focusing on Steam games. Another thing is that I would be improving the dashboard to make it look more complete and professional. I may need to adjust the layout, colors, charts, and filters so the dashboard is easier to read and more useful for users.

## **10. Conclusion**

This project successfully built a video game analytics pipeline using Python, Google Cloud Storage, BigQuery, and Looker Studio. The raw dataset was cleaned by using Python, stored in Google Cloud Storage, loaded into BigQuery, analyzed with SQL, and visualized in Looker Studio. The final dashboard helps users understand video game market trends better. Overall, this project helped me learn a lot about the full data analytics process pipeline. It also helped me learn how cloud tools can be used to store, process, analyze, and present data in a clear way.